Gauge flow as a unitary change of variables: invariance and truncation

On a finite lattice, gauge gradient flow is a diffeomorphism of the compact configuration space. Its unitary implementation preserves the spectrum and the norm of every transformed multiplication operator. Pointwise action reduction therefore does not reduce the corresponding worst-case norm. A changed local decomposition or projection may still change the bounds available to a proof, but it requires independent control of the generated terms and of any truncation or relative error. Neither unitary equivalence nor action decrease proves a mass gap.

1. Exact invariances

On $\mathcal{M} = G^{\mathcal{E}}$, the Wilson flow

$$\dot{V}_s(x,\mu) = -g^2\{\partial_{x,\mu} S_w(V_s)\}V_s(x,\mu), \qquad V_0 = U,$$

has a smooth all-time flow on the compact finite-dimensional manifold; $\Phi_t^{-1} = \Phi_{-t}$. Lüscher, arXiv:1006.4518v3, (1.4) and the following sentence (passage-level reading in the local companion), supplies the finite-lattice flow and action monotonicity. Let $\rho_t = |\det D\Phi_t|$ relative to Haar measure and define

$$(W_t\psi)(U) = \rho_t(U)^{1/2}\psi(\Phi_t(U)).$$

Change of variables proves W_t is unitary. Hence $W_t H W_t^*$ has exactly the spectrum, multiplicities and gap of H. Also

$$W_t M_f W_t^* = M_{f\circ\Phi_t}, \qquad \|f\circ\Phi_t\|_\infty = \|f\|_\infty,$$

because Φ_t is onto. Thus any criterion using only the spectrum of H_0 and the norm of ϕ is unchanged under *the same unitary conjugation* of the pair. This does not show invariance under an unrelated new splitting or under projection: flow can mix links and change the supports and hypotheses of a local theorem.

For the Kogut–Susskind Hamiltonian

$$H = \frac{\hbar c}{a}\left[\frac{g^2}{2}K + \frac{2}{g^2}V\right], \quad K = \sum_\ell(-\Delta_\ell), \quad V = \sum_p(N - \operatorname{Re}\operatorname{tr} U_p),$$

flow gives $0 \le V\circ\Phi_t \le V$ for $t \ge 0$, while its supremum is unchanged. It follows that $\langle\psi,(V\circ\Phi_t)\psi\rangle \le \langle\psi,V\psi\rangle$ for a fixed state ψ; if both state and observable are transformed consistently, their expectation is unchanged. In particular this is no proof that the physical ground-state measure has uniform local large-field tails. The corrected Agmon analysis controls integrated global excess only; the magnetic identities show that its simple Gibbs-comparison repair does not supply the missing local estimate.

2. The kinetic term and the free example

The electric operator becomes a second-order operator whose coefficients are determined by the pullback metric and the Jacobian. Flow mixes links: the transformed operator is generally nonlocal. The scale $\sqrt{8t}$ is a smoothing radius, not a strict finite interaction range. Bounds on the flow Jacobian do not by themselves prove relative operator-error bounds for a truncated Hamiltonian; the one-step distinction is explained below.

In a finite-volume regularized free abelian model in Coulomb gauge, a nonzero mode obeys $A_k \mapsto e^{-tk^2}A_k$ and the Jacobian is a constant. With a canonical momentum convention consistent with the Hamiltonian, the transformed quadratic coefficients acquire reciprocal factors e^{-2tk^2} and e^{+2tk^2}, so their product and oscillator frequency $\omega_k = c|k|$ are unchanged. This illustrates the exact unitary result. Treating the ratio of those coefficients as a Yarotsky smallness

parameter is not justified: Fourier modes are not the local site decomposition required by that theorem and a free mode's coordinate and momentum are unbounded. The former assertion that an arbitrary local criterion can always be improved without limit by conjugation is withdrawn.

One lattice step: coupling cancellation and local derivatives

For the stated Wilson normalization the g^2 in the flow equation cancels the $1/g^2$ in the action. At fixed dimensionless time, the finite-lattice vector field is therefore independent of g. Choosing $\sqrt{8t} = 2a$ gives $t/a^2 = 1/2$. The link-local vector field has bounded derivatives at fixed compact group and bounded lattice incidence. More explicitly, write its linearization in link coordinates as $\dot{J} = A(s)J$ with $A_{xy} = 0$ beyond a fixed interaction distance and row sums bounded by B, uniformly in the lattice size. Expanding the integral equation, a matrix element at graph distance d can receive a contribution only after at least d local factors, so

$$\|J_{xy}(t)\| \leq \sum_{n \geq d} \frac{(B|t|)^n}{n!}.$$

Here time and B are in lattice units, and any finite-range edge in A is one step of this graph. This elementary bound is uniform in volume and coupling under the stated derivative and frame bounds; its constants depend on the group and local geometry. It controls the first flow derivative, not automatically the inverse derivative, Jacobian potential, unbounded kinetic form or full Hamiltonian truncation. The former claim Ce^{-cK} for the relative operator error and the assertion that $K = 3$ suffices are withdrawn: no constants or relative-form proof were provided. Compactness also does not itself prove normalized large-field rarity or multiscale stability.

3. What an RG step must add

Suppose a scale change constructs effective operators $H^{(n)}$ at $\ell_n = 2^n a$, with a specified low-energy subspace, generated interactions and a spectral comparison. Relative Schur bounds can ignore a common scalar shift but must control the state-dependent remainder and the complementary energy denominator; see the Feshbach reduction, including its ultraviolet correction. A candidate truncation must be compared with the exact transformed operator in a suitable domain and relative-form norm, with constants uniform in volume and with enough control across the cutoff trajectory. One-loop running can estimate how many doublings might be needed, but it is not a proof of the effective action or its arrival in a gapped region.

A state-relative bound, for example

$$\langle \psi, \phi\psi \rangle \leq \epsilon \langle \psi, (H - E_0)\psi \rangle + \eta \|\psi\|^2$$

on a specified low-energy domain, could exploit more information than a supremum norm. Such a bound, the relevant probability estimates, and stability after projection are independent obligations. The blocking analysis gives conditional surface-to-gap comparisons; it does not prove that projection blocking is always worse at every coupling.

4. Consequence for STATE

Exact flow preserves spectra and transformed norms. The free cancellation and the failure of the global Agmon bound to yield a local state estimate make the missing ingredients explicit: controlled generated interactions, relative errors and physical spectral transfer. No general no-go theorem for all norm-based methods, all reblockings or all constructive routes is claimed.